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## **Mitigating Foam and Maximizing Throughput in Critical Sour Gas Sweetening Units**

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### **Abstract**

One of the largest critical sour facilities in PDO operates two gas sweetening units (GSUs), each designed for 4.7 MMSCMD. Both units faced persistent foaming and high Sulfinol-X losses, even while operating within design parameters. This paper presents technical study, process enhancements, and practical solutions implemented to resolve these challenges. The primary objective was to eliminate foaming, reduce chemical consumption, improve plant stability, and prevent gas production deferment, while establishing a proactive troubleshooting protocol.

The paper outlines a structured approach to mitigating excessive foaming and solvent losses in critical sour GSUs. A thorough root cause analysis was conducted to identify the root causes of foaming. As a result, antifoam usage was fully eliminated, solvent circulation was optimized, and overall system stability improved. These actions led to increased gas throughput, substantial reductions in chemical and energy consumption, and complete avoidance of deferment. Following resolution of the foaming issue, a proactive PI-based monitoring and response tool was developed to enable early detection and timely intervention.

**Keywords:** Amine foaming, Gas sweetening, Sulfinol-X, Process optimization, Sour gas, Solvent circulation, Foaming control

### **Introduction**

Foaming is a common and recurring challenge in gas sweetening units (GSUs), particularly in facilities processing critical sour gas (Fatemi et al., 2018). It can significantly impact unit performance by causing solvent losses, reduced absorption efficiency, poor gas-liquid separation, high pressure drops, and ultimately, production deferment. Foaming results from the presence of foam promoting impurities in the Methyldiethanolamine (MDEA) solvent (Shahouni et al., 2024).

Foaming in GSUs can be triggered by several factors - 1) The degradation products formed from amines by chemical reaction with oxidants and foreign acids or by exposure to high temperatures may be foam promoting and their presence should therefore be prevented (Alhseinat et al., 2015). 2) Foaming is clearly linked to ingress of liquid hydrocarbons into the amine system. Liquid hydrocarbons may enter the system

via entrainment in the feed gas or condensation may occur in the absorber because either the lean solvent temperature is below the feed gas dew point or removal of acid gas raises the dew point of the treated gas above the lean solvent temperature (Chen et al., 2022). 3) By the nature of the process, formation of Iron Sulfide (FeS) particulates occurs particularly during the initial start-ups after maintenance shutdowns, when the presence of iron oxides is more likely (Hunter et al., 2008). 4) Corrosion inhibitors, particularly the filming amine types, are by their nature surface active agents and may be severe foam promoters. They can enter the amine system via the feed gas or with the solvent make-up water which is often boiler feed water. The use of filming amine type corrosion inhibitors has not been found to be consistently effective in amine systems (Oshinowo et al., 2022, Shahouni et al., 2024). 5) Excessive use of antifoam can in fact lead to foaming. Many anti-foam formulations include an emulsifier to allow the rapid distribution of the active agent through the amine system. Continued use of anti-foam can build up the background level of emulsifier and hence create foaming (Al Ali et al., 2022). 6) Foreign materials originating from related equipment such as lubricants from compressors, grease (containing soap bases) from plug valves, conservation oils/greases from new equipment, filter elements made of cotton material may all promote foaming. 7) Sometimes rapid changes of gas flow can provoke a foaming incident, particularly in systems where other foam promoting factors as above are already present (Shahouni et al., 2024).

In systems using Sulfinol-X, which combines physical and chemical absorption, maintaining solvent integrity and proper operating conditions is critical to avoiding such issues. At one of Petroleum Development Oman's (PDO) largest critical sour gas facilities, both parallel GSUs (each rated for 4.7 MMSCMD) experienced persistent foaming and high Sulfinol-X losses despite operating within design parameters. The complexity and severity of the issue demanded a comprehensive technical study to identify the root causes and implement lasting solutions.

This paper presents a systematic approach to resolving foaming and solvent loss issues in these units. It details the process troubleshooting, optimization trials, and operational changes that led to improved unit stability, elimination of antifoam usage, and increased throughput. The work also introduces a proactive monitoring tool designed to detect early signs of foaming and enable timely corrective actions.

## Process Overview

Production gas from the R field is directed to two parallel gas sweetening units (GSUs). The GSUs at R Station are designed in a 2 × 50% train configuration, with a combined capacity to sweeten approximately 9.4 MMSCMD of peak sour gas. The sour gas flow is distributed between the two treatment trains for balanced operation.

The acid gas separated in the regenerator overhead condenser is routed to the acid gas compression train and subsequently dehydrated in a TEG dehydration unit before being injected into the reservoir.

Meanwhile, the treated gas from the GSUs flows to the Mercaptan Sweetening Unit (MSU), where residual mercaptans are removed. This is followed by mercury removal, hydrocarbon dewpointing, and finally, export of the sweetened gas.

## Background on Gas Sweetening

Gas sweetening is the process of removing acidic components such as hydrogen sulfide (H<sub>2</sub>S), carbon dioxide (CO<sub>2</sub>), carbonyl sulfide (COS), and mercaptans (RSH) from sour gas streams. At the R Station gas sweetening facility, this is achieved by contacting sour gas with a formulated solvent, Sulfinol-X, which has a strong affinity for acidic gases and ensures the production of sweet export gas. Figure 2 represents a typical Gas sweetening unit:

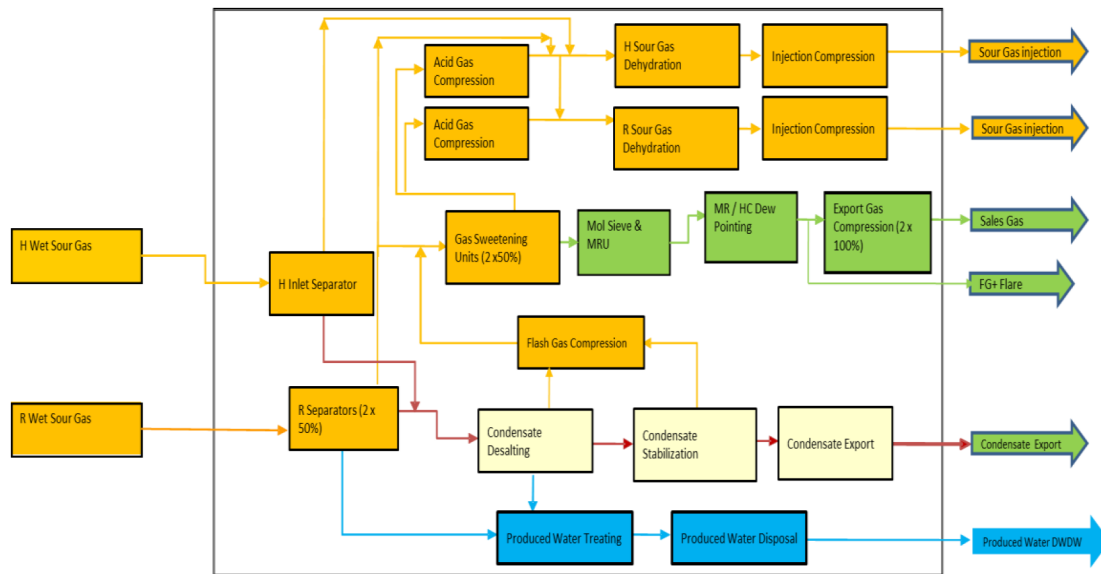


Figure 1—R Station Process overview

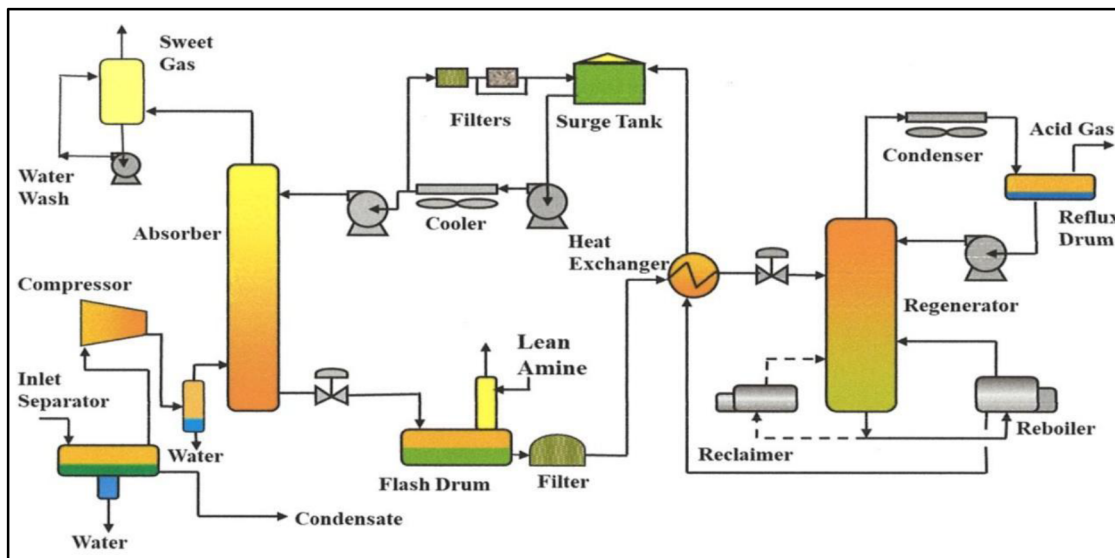


Figure 2—Typical Gas sweetening unit

Sulfinol-X is a proprietary Shell Global Solutions solvent that combines the advantages of chemical and physical absorption. It is typically composed of:

- 40-45% amine (MDEA or secondary amines such as piperazine),
- 35-40% sulfolane (organic physical solvent),
- 15-25% water.

This formulation enables effective removal of  $H_2S$ ,  $CO_2$ ,  $COS$ , and  $RSH$ , while offering several operational benefits:

- Higher capacity due to reduced solvent circulation rates.
- Lower energy requirements and smaller equipment sizing.

- Reduced solvent degradation compared to conventional amine solvents.
- Lower corrosion risk and extended solvent life without frequent replacement.

The **R Station GSUs** were designed with a dual-train configuration ( $2 \times 50\%$ ) and a peak processing capacity of approximately 9.4 MMSCMD sour gas. Each train utilizes Sulfinol-X containing:

- 43 wt% MDEA (tertiary amine for complete  $H_2S$  and partial  $CO_2/COS$  removal),
- 7 wt% piperazine (secondary amine for accelerated  $CO_2$  and  $COS$  absorption),
- 15 wt% sulfolane (enhances RSH and  $COS$  removal),
- 35 wt% water.

This enhanced solvent blend provides superior treating efficiency, reduced degradation (no oxazolidone formation), and significant operational cost savings compared to conventional amine

## Methodology

To effectively resolve the persistent foaming issues in the gas sweetening units, a comprehensive and structured methodology was implemented. This approach combined detailed root cause analysis, process optimization trials, operator engagement, and digital monitoring tools to ensure sustainable performance improvement.

A detailed root cause analysis (RCA) was conducted to systematically identify all potential sources and indicators of excessive foaming. This analysis included a thorough review of historical process data, chemical usage patterns, operating conditions, and solvent quality. Gas samples were analyzed to measure total sulfur,  $H_2S$ , and  $CO_2$  concentrations, providing a chemical profile to validate and support the findings.

As part of this RCA, a complete list of foaming indicators was developed, each associated with its specific root cause and recommended corrective action. These indicators ranged from solvent circulation anomalies and lean/rich amine loading imbalances to signs of contamination, poor separation efficiency, and unexpected pressure drops. This comprehensive table of indicators, causes, and actions was compiled and shared with the operations team as part of a standard operating procedure (SOP) guideline.

## Results and Discussion

Foaming is recognized by one or more of the following symptoms:

- High, fluctuating and rising pressure drop over the absorber and/or regenerator columns; typically, the pressure drop starts to oscillate, the oscillations grow larger and then suddenly the pressure drop increases dramatically indicating heavy foaming and possibly carryover. Intervention should take place as soon as the early warning signs of oscillating pressure drop is observed.
- Loss of level in the bottom of the absorber and/or regenerator column; typically happens simultaneously with the increasing pressure drop.
- Carry-over of solvent in the treated gas K.O. vessel.
- High level in the regenerator reflux vessel because of carry-over.
- Liquid carry-over to sulphur recovery unit feed knockout pots.
- Rapidly increasing levels of  $H_2S$  in the treated gas.

The following measures are suggested for immediate response to a foaming incident. It is important that once these steps are taken the operator then investigates the underlying cause of the foaming.

## Absorber

- Reduce the lean solvent flow to the foaming absorber in one step. This will allow the foaming solvent to drain to the bottom of the column (the level will rise). After a short period (15 min) the solvent flow to the absorber can be increased again in small steps.
- If a bypass is available the feed gas can be partially bypassed to reduce the flow through the column which will also help recovery from foaming.

## Regenerator

- At severe foaming, reduce the reboiler steam by 30% in one step. This will allow the foaming solvent to drain to the bottom of the regenerator (level will rise). After a short period (say 10 minutes) the steam can be increased in small steps to its normal rate.

## Anti foam injection

- If experiencing continuing foaming, a limited shot of anti-foam can be injected in the system (1 to 5 ppm weight on inventory of undiluted AFA (= Active Ingredients) during a 5 minutes period. Number of shots of AFA should be limited to 1 per shift as maximum.

## Root Cause Analysis of Foaming in the GSU

Table 1 summarizes the key indicators of foaming observed in sour gas sweetening units, matches each indicator to its likely root cause, and defines the corresponding operator action. This format can be incorporated into the SOP, and it mirrors how the PI monitoring dashboard triggers visual (traffic-light) alerts.

The regenerator column bottom temperature was observed to be low which was attributed to high water content in lean Sulfinol. When water content in lean sulphinol increases the solvent becomes diluted in the active components which are sulphonale and amine. This leads to less absorbance of  $H_2S$  and  $CO_2$  in the absorber and more is carried to the regenerator. With high water content vaporization starts earlier at lower temperature and more energy is spent on evaporation of water instead of sulphonale rich solvent. The column thus operates at a lower bottom temperature leading to potential foaming. Increasing hot steam flow to the regenerator will raise the bottom temperature of the regenerator and the evaporation rate which strips more water overhead, thus lowering the water content of the lean sulphinol leaving the regenerator.

High  $H_2S$  content in the feed gas or low circulation rates lead to high Sulfur loading in the solvent. This changes the physical properties of the solvent. The surface tension of the solvent drops leading to easier bubble formation and the viscosity rises leading to slow bubble collapse making the solvent more prone to foaming. High Sulfur content also promotes the thermal and oxidative degradation of amines in the sulphinol mixture which acts as surfactants that form stabilized foam films. Sulfur compounds can lead to corrosion by product formation like iron sulphide, iron carbonate etc., that enter the solvent and act as foam stabilizers forming rigid films around bubbles aggravating foaming. All these effects of high Sulfur content led to operational issues like high gas throughput and overloaded regenerator.

When the feed gas is not sufficiently dehydrated before entering the absorber or the absorber is operating too close to the dew point condensation occurs in the absorber. Condensation of gases in the absorber leads to foaming. The condensed liquids like hydrocarbons act as surfactants leading to lowering of surface tension of sulphinol and, hence stabilizing the bubbles. The condensed films on packing/trays interfere in gas-liquid contact area lowering the mass transfer process. Since the contact efficiency drops foaming interferes with absorption. The condensate reaching the HP flash gas compressor leads to high discharge temperature, vibration or even liquid slugging (Rohani et al., 2017). Foaming pushes the solvent into gas

stream leading to solvent losses. Condensation can be prevented by dew point margin control, dehydration, solvent temperature optimization and proper separator maintenance. In our study the temperature of the stream was set to the optimum temperature.

**Table 1—Summary of Foaming Triggers, Root Causes, and Recommended Control Actions for Sour- Gas Sweetening Units**

| Sr No | Indicator                                                        | Cause                                               | Actions                                                |
|-------|------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| 1     | Regenerator column bottom temperature (Low)                      | High water in lean Sulfinol                         | Close make up water valve, and increase hot steam flow |
| 2     | Treated gas Total Sulphur (High)                                 | High water in lean Sulfinol                         | Close make-up water valve, and increase hot steam flow |
|       |                                                                  | Low Sulfinol circulation rate                       | Increase Sulfinol circulation rate                     |
|       |                                                                  | High gas rate compared to Sulfinol circulation rate | Reduce inlet gas flow rate into GSU                    |
|       |                                                                  | Lean Sulfinol high temperature                      | Start Lean Sulfinol cooler fan                         |
| 3     | HP flash gas compressor aftercooler discharge temperature (High) | Gas condensation in absorber column                 | Change the set point of stream to optimum temperature  |
| 4     | Treated gas total Sulphur (Low)                                  | High Sulfinol circulation rate                      | Reduce Sulfinol circulation rate                       |
| 5     | Inlet flow rate to GSU (High)                                    | High station header pressure                        | Maintain station header pressure                       |
|       |                                                                  | GSU outlet valve high opening                       | Reduce the opening of GSU outlet valve                 |
| 6     | Fresh Sulfinol level in flash vessel (Low)                       | Low level in fresh Sulfinol vessel                  | Makeup Sulfinol in flash vessel                        |
| 7     | Reflux drum level increasing without makeup                      | More DM water in lean Sulfinol                      | Increase reflux flow rate and hot steam flow           |

High circulation rate of sulphinol increases the amount of sulphinol and sour gas contact. This leads to more absorption of  $H_2S$  and hence a low Sulphur content in the treated gas. High circulation on the other hand may strain the system, increase energy cost, increase the thermal and oxidative stress on sulphinol. Hence great care must be taken to match the circulation rate to the absorber design and regenerator capacity to avoid inefficiencies.

The high pressure of sour gas entering the sweetening unit, also known as the high station header pressure, leads to increased gas solubility in sulphinol, higher gas flow rate through the absorber, increased solvent loading and regenerator duty, potential equipment stress and operational impacts. Foaming risks increase due to higher gas velocity.

In a gas sweetening unit, the fresh Sulfinol level in the flash vessel plays a critical role in foaming. The flash vessel is designed to remove dissolved gases such as  $H_2S$  and  $CO_2$  from the rich Sulfinol before it enters the regenerator. When the fresh Sulfinol level is too high, the available space for gas disengagement is reduced, causing gas bubbles to form more violently within a smaller liquid volume. Fresh Sulfinol, being highly active, contains a high concentration of absorbed acid gases, which rapidly evolve during flashing, intensifying bubble formation. This vigorous bubbling promotes foam generation, which can carry over into downstream equipment, disrupt regenerator operation, and lead to operational issues such as liquid entrainment, pressure fluctuations, and decreased gas separation efficiency. Therefore, maintaining an optimal liquid level is essential to control foaming and ensure stable operation of the gas sweetening unit.

In a gas sweetening unit, the reflux drum (overhead accumulator of the regenerator) is designed to separate the condensed reflux solvent from acid gases and return it to the column to maintain stable stripping and cooling. If the reflux drum level increases without makeup (i.e., without corresponding solvent balance adjustment), it often indicates the accumulation of contaminants such as hydrocarbons, degradation products, or condensed heavy ends. These impurities reduce solvent quality and stability, making the circulating Sulfinol or amine more prone to foaming. An increased reflux drum level also means a higher volume of liquid is being returned to the regenerator, which can upset the column's liquid-to-vapor balance, creating entrainment and turbulent flow that further promotes foaming. Once foaming begins, it reduces effective gas-liquid contact in the regenerator, impairs acid gas stripping efficiency, and can lead to solvent carryover into the acid gas stream. Thus, an unchecked increase in reflux drum level without proper makeup or purge directly contributes to foaming and destabilizes sweetening unit operation.

### **Foaming PI process book proactive monitoring tool**

To ensure continuous monitoring and proactive response, a **PI ProcessBook** tool was developed. This tool was designed specifically for control room operators to visualize all key foaming indicators in a user-friendly interface. Key features of the PI tool include:

- Real-time display of critical process parameters associated with foaming.
- Traffic light indicators for quick visual assessment of system health.
- Predefined actions for each alert condition, enabling prompt and accurate operator response.
- Integration with SOP content for easy reference during abnormal conditions.

Table 2 presents the operating key foaming indicators of GSU-1 and GSU-2 compared against their respective target ranges using the PI ProcessBook tool. The indicators include reboiler temperature, sulfur content, inlet flow, water levels, and compressor set points.

Table 2 highlights critical areas where process adjustments are necessary to maintain stable operation and prevent potential foaming or efficiency loss in the gas sweetening units. Overall, the developed tool significantly improved the responsiveness of the operations team, allowing early detection and rapid mitigation of potential foaming issues. As a result, foaming incidents were reduced, solvent losses were minimized, and overall system stability improved.

This study clearly demonstrates that a combined strategy of process-parameter tuning, dedicated operator engagement, and proactive real-time monitoring can effectively mitigate foaming, significantly cut costs, and boost the operational reliability of gas sweetening units (GSUs).

Before the intervention, the units experienced 10-20 foaming incidents per day. After implementing the optimized Sulfinol-X circulation, eliminating antifoam usage, and deploying the PI Process Book tool, the frequency dropped to an average of 0-2 incidents per day. This multidimensional intervention underscores the importance of diagnosing root causes rather than relying solely on short-term fixes.

Table 2—Operating Key Foaming Indicators of GSU-1 and GSU-2 using PI ProcessBook Tool

| Indicator                                   | Action                                                     | Target Range                   | GSU-1          | GSU-2          |
|---------------------------------------------|------------------------------------------------------------|--------------------------------|----------------|----------------|
| Reboiler Bottom T                           | Control steam flow to reboilers                            | 133.5 – 134.5 °C               | ● 134.56 °C    | ● 134.55 °C    |
| High Total Sulfur                           | Increase Circulation Rate                                  | Below 20 ppmv                  | ● 11 ppmv      | ● 10 ppmv      |
| Low Total Sulfur                            | Reduce Circulation Rate                                    | Below 3 ppmv                   | ● 11 ppmv      | ● 10 ppmv      |
| HP Flash Gas Compressor after cooler TIC SP | Maintain SP below 59                                       | Below 59 °C                    | ● 58 °C        | ● 58 °C        |
| Maximum GSU Inlet Flow                      | Control GSU Inlet Flow                                     | GSU1 = 4.6, GSU2 = 3.6 MMSCM/D | ● 3.95 MMSCM/D | ● 3.61 MMSCM/D |
| High Water in Lean Sulfinol                 | Maintain reflux vessel level by controlling 40-FCV-122/222 | Above 30%                      | ● 23.55 %      | ● 28 %         |
| High DM Water Makeup Flow                   | 40-FCV-122/222 opening should be less than 3%              | Below 3%                       | ● 0 %          | ● 37 %         |
| Low V-4023 Level                            | Makeup sulfinol in flash vessel till 20% level             | Above 10%                      | ● 7.07 %       | ● 15.94 %      |

Figure 3 and 4 show the foaming frequency in one of the GSUs before and after implementing the changes.

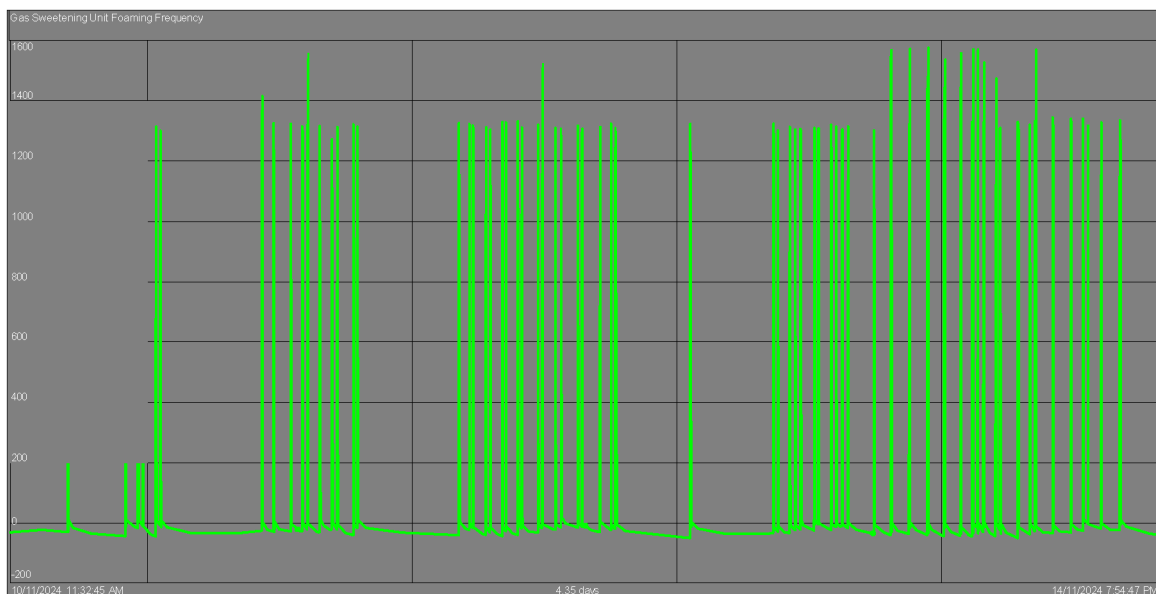


Figure 3—GSU Foaming Frequency Before Implementing Changes

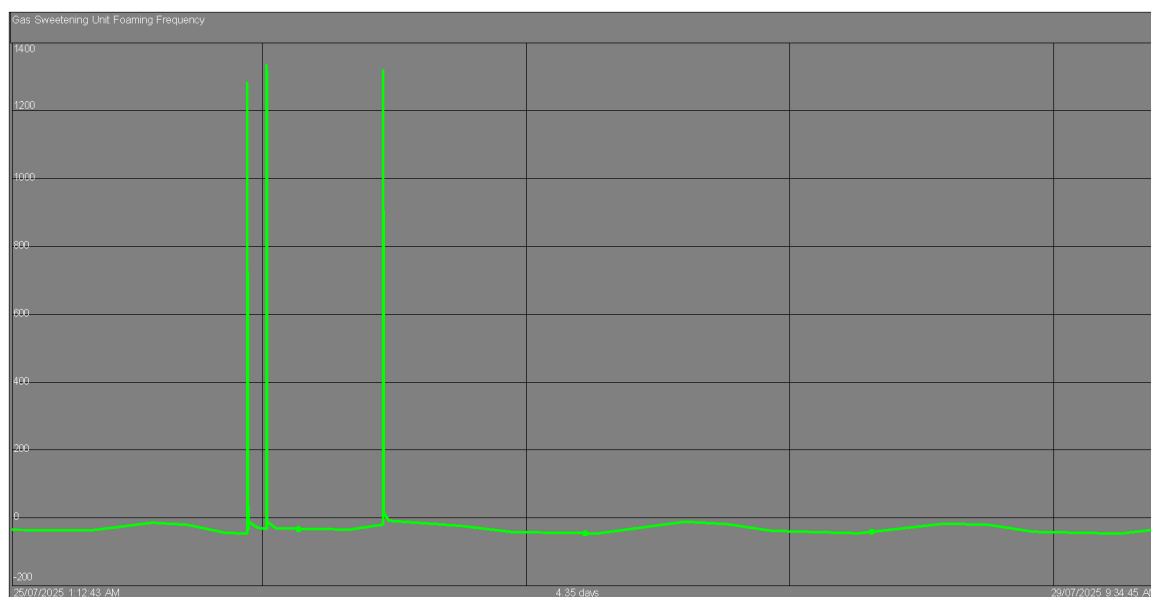


Figure 4—Current GSU Foaming Frequency After Implementing Changes

## Conclusion

The integrated optimization initiative implemented at R Station led to substantial operational and financial benefits:

- Foaming incidents dropped to near-zero levels, with 0 antifoam injections per day, eliminating solvent carryover and stabilizing GSU hydraulics.
- Export capacity increased to 5.6 MMSCMD, exceeding the design rate of 5.0 MMSCMD.
- Solvent circulation was reduced from 28,000 m<sup>3</sup>/day to 20,000 m<sup>3</sup>/day, maintaining sweet gas quality while significantly lowering energy demand.
- Annual savings included approximately US \$30,000 in antifoam chemicals and 10,500 MWh of energy (= US \$0.8 million).
- Recovered 22,500 m<sup>3</sup> of condensate annually, avoiding deferred volumes valued at about US \$5.5 million/year.

These results confirm that systematically combining targeted process adjustments, plant-level troubleshooting, operator-focused procedures, and live monitoring tools deliver a step-change in GSU effectiveness. The initiative not only enhanced chemical efficiency and plant reliability but also enabled optimized throughput well beyond initial capacity.

By empowering the operations team with a clear SOP-aligned foaming control matrix and live-alert monitoring dashboard, the unit achieved a consistent, sustainable operating discipline—demonstrating a blueprint that can be adopted by other critical sour gas facilities aiming to eliminate foaming and drive value.

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